4.3. PRODUCTION AND INVENTORY

What is learned in this exercise

- Build a model using successive versions.
- Delay SMOOTH and function STEP.
- Customizing graphs to compare variables.
- Define an external table for non-linear relationship.
- Study the effects of a policy using a model.

A manufacturing firm has experienced chronic instability in its inventory stock and production rate. Discussions with management and workers on the factory floor have revealed the following information:

1. The firm supplies its customers from a stock of finished inventory, which is generally sufficient to meet orders as they are received.

2. Desired production is determined by anticipated (forecasted) shipments, modified by a correction to maintain inventory at the desired stock.

3. The firm forecasts shipments by averaging past orders over an eight-week period as a way of smoothing out any noise or lumpiness in demand.

4. The firm tries to correct discrepancies between desired and actual inventory in eight weeks.

5. Desired inventory is four weeks' worth of anticipated shipments.

The firm's actual production rate is assumed to equal desired production. The initial value for inventory is assumed to be equal to desired inventory, and the initial value of the average order rate is assumed to be equal to the shipment rate. As a result, the model will begin in an initial equilibrium regardless of the initial customer order rate.

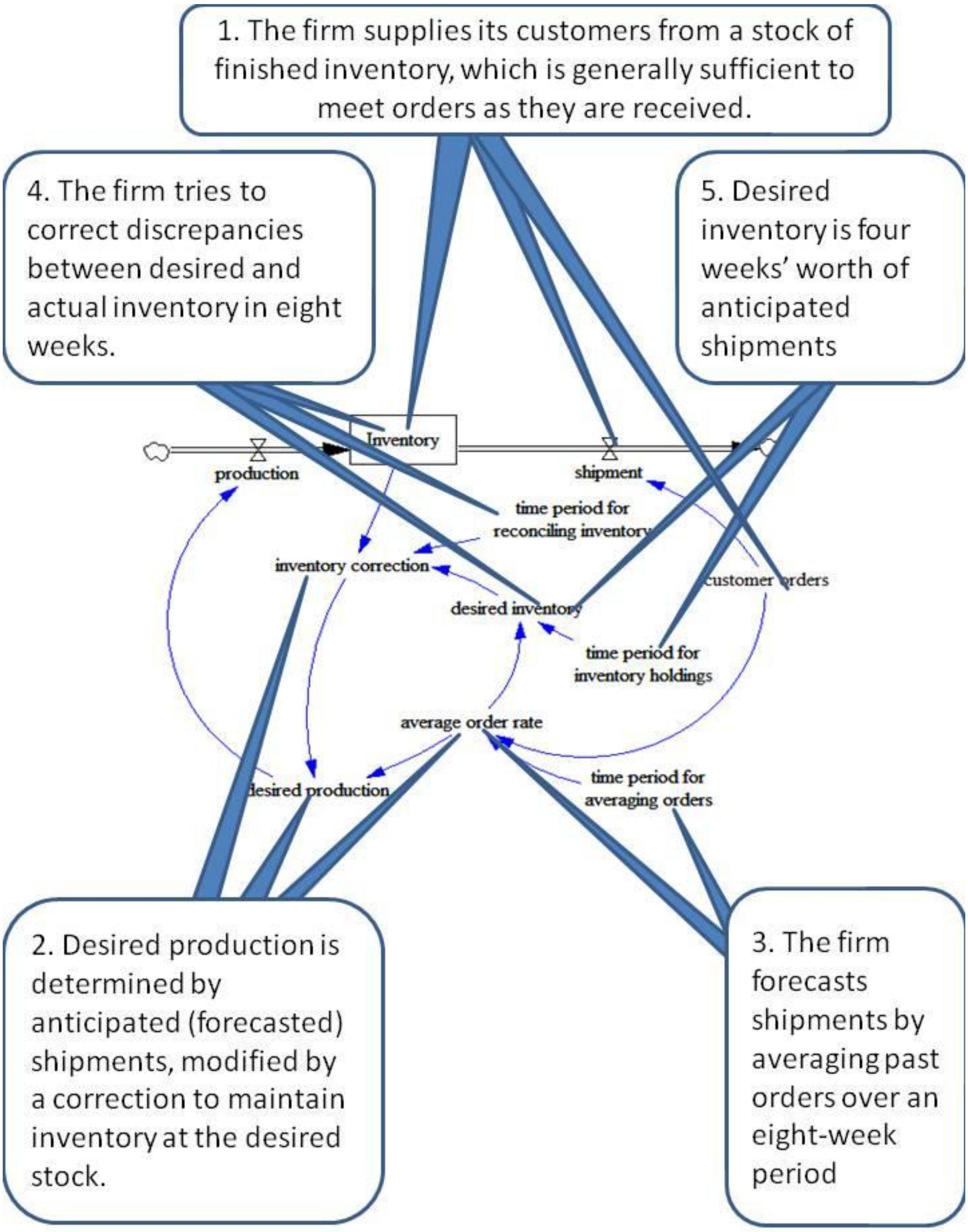
In this model customer orders will equal a constant of 1,000 units/week until the period 10, when orders step up by 10%, and then remain at the higher rate.

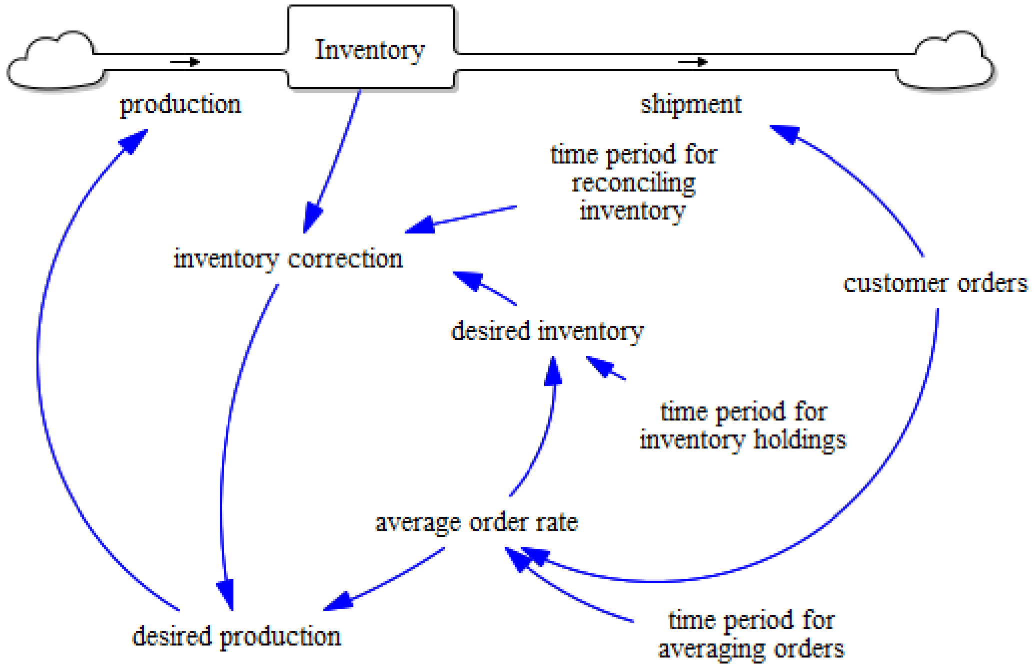
Preliminary question

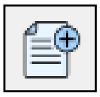
Before making the model, draw the behaviour you expect the model will have when tested with the 10% step increase in orders (starting from equilibrium). Include the behaviour of orders, desired production, production, inventory, and desired inventory. Explain in one paragraph the reasons for the behaviour you expect.

From text to the diagram

Each of the sentences that explain the rules that the company follows can be translated as a stock and flow diagram.







Model 1

Click File-New model, and define:
INITIAL TIME = 0 FINAL TIME = 100 TIME STEP =1 Units for time = Week

Equations:

(01) average order rate=SMOOTH(customer orders, time period for averaging orders)
Units: units/Week
The firm forecast shipments by averaging past orders over an eight-week period (using an exponentially-weighted moving average) as a way of smoothing out any noise or lumpiness in demand.

(02) customer orders=1000*(1+STEP(0.1,10))
Units: units/Week
In this model customer orders will equal a constant of 1,000 units/week until period 10, when orders step up by 10%, and then remain at the higher rate.

(03) desired inventory=average order rate*time period for inventory holdings
Units: units
Desired inventory is four weeks' worth of anticipated shipments.

(04) desired production=average order rate+ inventory correction
Units: units/Week
Desired production is determined by anticipated shipments, modified by a correction to maintain inventory at the desired level.

(05) Inventory= production-shipment
Initial value : 4000
Units: units
The initial inventory is four weeks' worth of shipments (1,000 x 4 = 4,000)

(06) inventory correction= (desired inventory-Inventory)/time period for reconciling inventory
Units: units/Week
The firm tries to correct discrepancies between desired and actual inventory in eight weeks.

(07) production=desired production
Units: units/Week

(08) shipment= customer orders
Units: units/Week

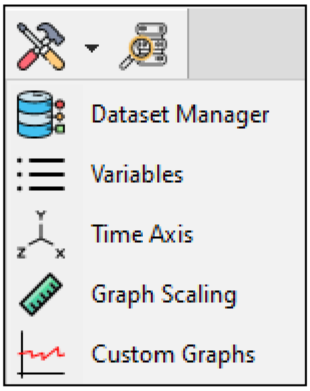
(09) time period for averaging orders= 8
Units: Week
Time to average order rate

(10) time period for inventory holdings= 4
Units: Week
Desired inventory coverage

(11) time period for reconciling inventory= 8
Units: Week
Time to correct Inventory

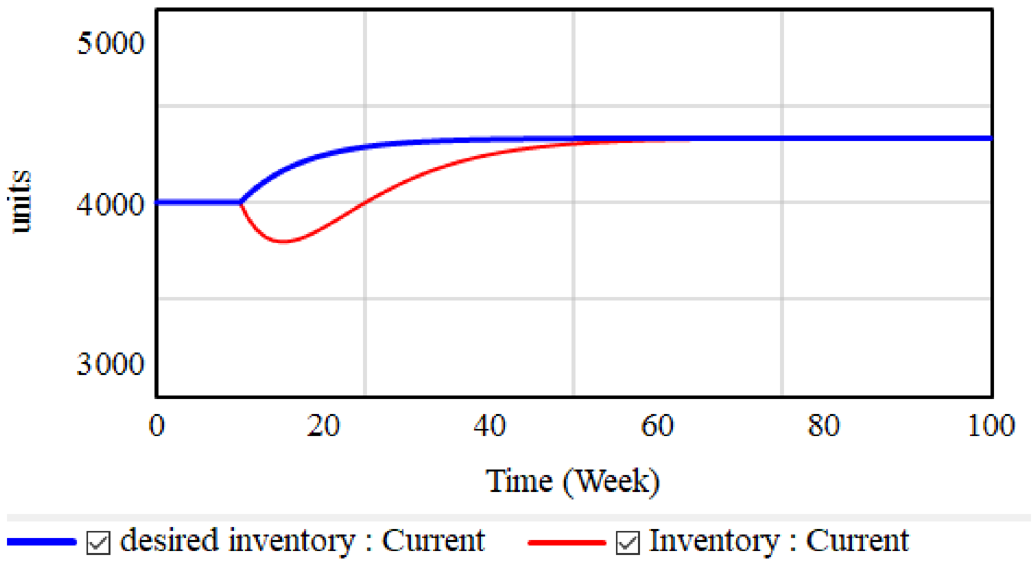
Simulate

To see more than one variable in the same graph go to Control panel – Custom Graphs – New and select the variables, share the scale, and define the Y-min and Y-max values. OK.



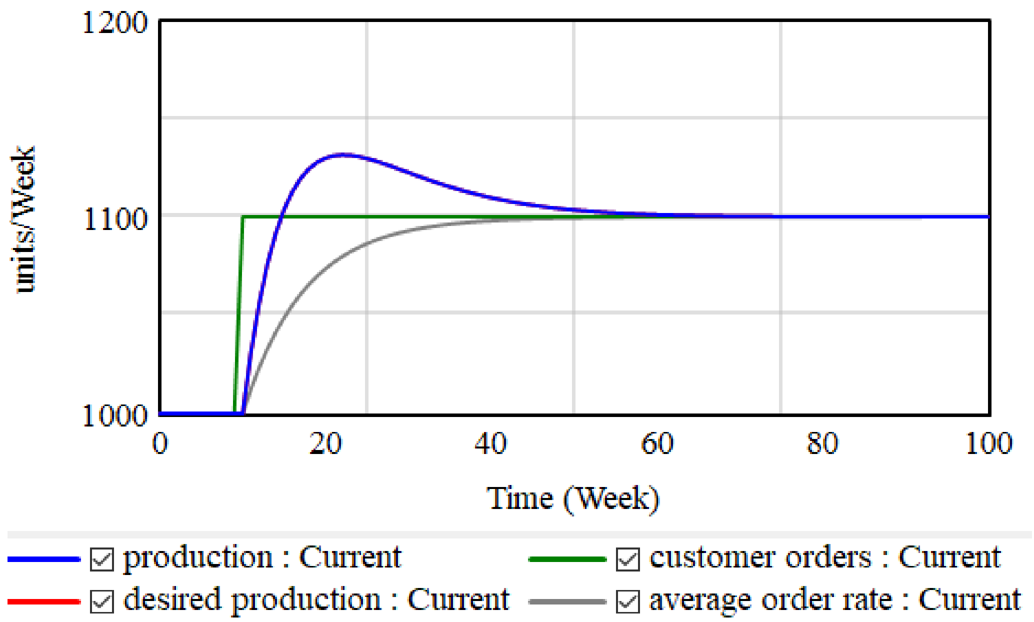
Inherit scale	Variable	Dataset	Label	Line thickness	Pattern	Color	Units	Y-min	Y-max
	desired inventory			3	Solid line			2800	5200
☑	Inventory			2	Solid line				

Then select and click Display.



Define now a new graph:

Inherit scale	Variable	Dataset	Label	Line thickness	Pattern	Color	Units	Y-min	Y-max
	production			2	Solid line			1000	1200
☑	desired production			2	Solid line				
☑	customer orders			2	Solid line				
☑	average order rate			2	Solid line				



Questions

a. After running the model compare the actual behaviour with your expected behavior for each of the variables, explaining the reasons for the actual behaviour and why it differs from your expectations (if it does).

b. Shorten 'time period for reconciling inventory' to four weeks.

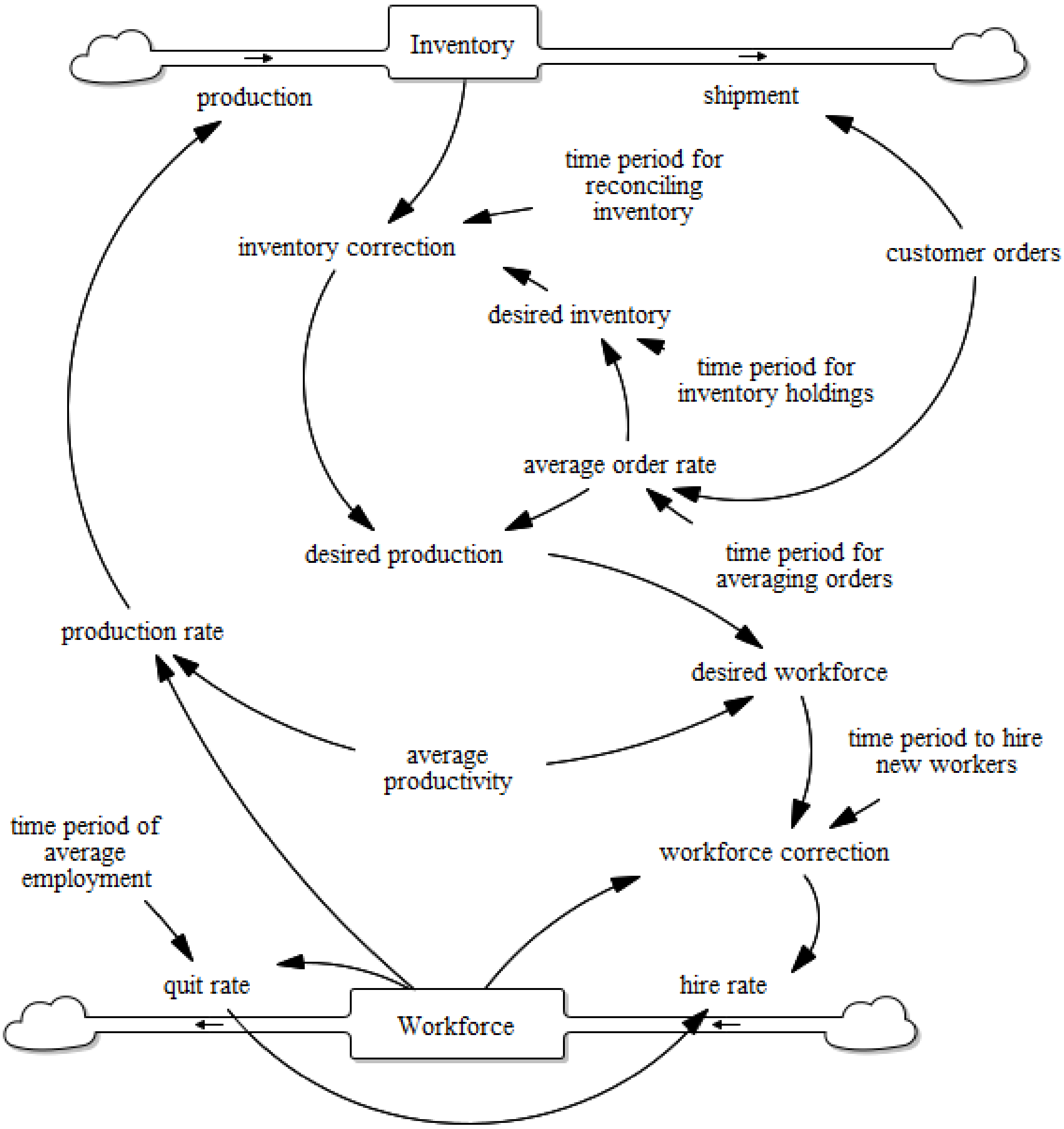
- Does the behaviour mode change?
- Does the timing of the behaviour change?
- It is more or less stable?
- Explain why or why not.

Model 2

Noting that the current model assumes actual production equals desired production, you return to the company to check on the validity of that assumption. Further discussions reveal that while the firm has ample physical plant and equipment, labour cannot be hired and trained instantaneously. In fact, it takes approximately 24 weeks to advertise for, hire, and train new workers. The firm has a no-layoff policy, and workers stay with the firm an average of 50 weeks (one year).

The firm's hiring policy is to replace those workers who quit, modified by a correction to bring the actual workforce into balance with the desired workforce. Because workers must give two weeks' notice, there is no significant delay between quits and replacement hiring. Because of the delays in hiring new people, it takes 24 weeks to make corrections to the workforce. The desired workforce is determined by desired production and average productivity, which is equal to 20 units per worker per week and is quite constant over time. Finally, union rules prevent the use of overtime or undertime.

Note in this model that the recruiting, hiring, and training delays have been aggregated together into the 'time period to hire new workers'



(01) average order rate=SMOOTH(customer orders, time period for averaging orders)
Units: units/Week
The firm forecast shipments by averaging past orders over an eight-week period as a way of smoothing out any noise or lumpiness in demand.

(02) average productivity= 20
Units: units/person/Week

(03) customer orders= 1000*(1+STEP(0.1,10))
Units: units/Week
In this model customer orders will equal a constant of 1,000 units/week until period 10, when orders step up by 10%, and then remain at the higher rate.

(04) desired inventory=average order rate*time period for inventory holdings
Units: units
Desired inventory is four weeks' worth of anticipated shipments.

(05) desired production=average order rate+ inventory correction
Units: units/Week
Desired production is determined by anticipated shipments, modified by a correction to maintain inventory at the desired level.

(06) desired workforce=desired production/average productivity
Units: person

(07) hire rate=quit rate+ workforce correction
Units: person/Week

(08) Inventory= production-shipment
Initial value: 4000
Units: units
The initial inventory is four weeks' worth of shipments (1,000 x 4 = 4,000)

(09) inventory correction=(desired inventory-Inventory)/time period for reconciling inventory
Units: units/Week
The firm tries to correct discrepancies between desired and actual inventory in eight weeks.

(10) production= production rate
Units: units/Week

(11) production rate= Workforce*average productivity
Units: units/Week

(12) quit rate= Workforce/time period of average employment
Units: person/Week

(13) shipment= customer orders
Units: units/Week

(14) time period for averaging orders= 8
Units: Week
Time to average order rate

(15) time period for inventory holdings= 4
Units: Week
Desired inventory coverage

(16) time period for reconciling inventory= 8
Units: Week
Time to correct Inventory

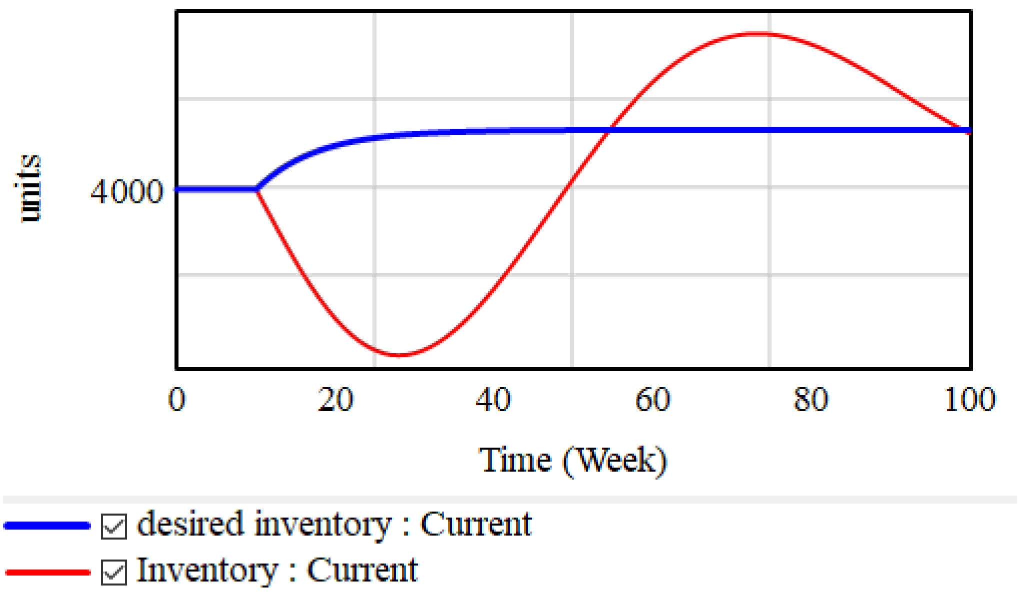
(17) time period of average employment= 50
Units: Week
Average duration of employment

(18) time period to hire new workers=24
Units: Week
Time to correct workforce

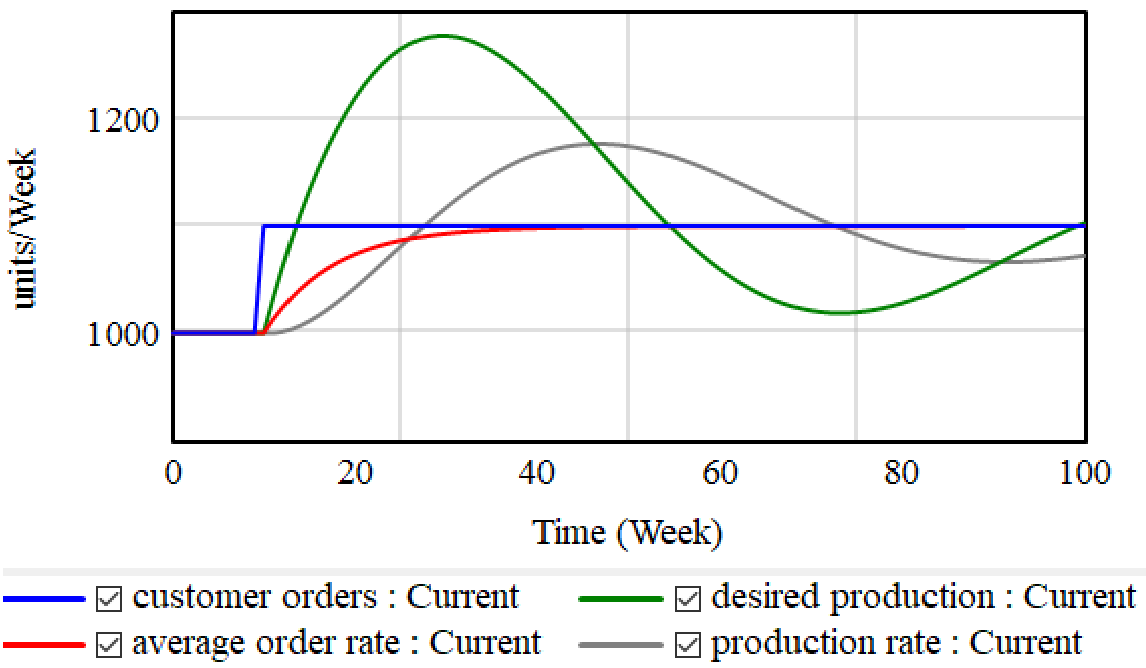
(19) Workforce= hire rate-quit rate
Initial value: 50
Units: person

(20) workforce correction=(desired workforce-Workforce)/time period to hire new workers
Units: person/Week

Inherit scale	Variable	Dataset	Label	Line thickness	Pattern	Color	Units	Y-min	Y-max
	desired inventory			3	Solid line			2800	5200
☑	Inventory			2	Solid line				



Inherit scale	Variable	Dataset	Label	Line thickness	Pattern	Color	Units	Y-min	Y-max
	customer orders			2	Solid line			900	1300
☑	average order rate			2	Solid line				
☑	desired production			2	Solid line				
☑	production rate			2	Solid line				



Questions

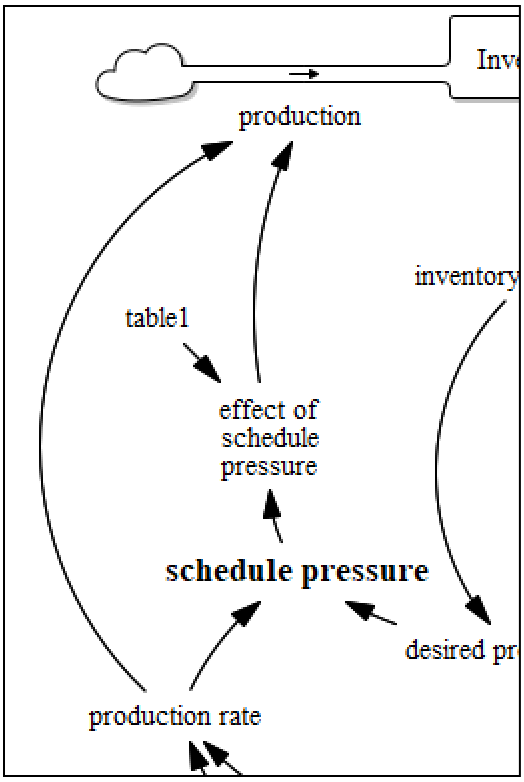
Managers control the ‘time period for reconciling inventory’, which represents how aggressively the firm tries to correct inventory discrepancies. A long ‘time period for reconciling inventory’ implies management is willing to allow a large inventory gap, while a short ‘time period for reconciling inventory’ implies swift corrections to eliminate a gap and bring inventory in balance with desired inventory rapidly.

a. To stabilize the system, should ‘time period for reconciling inventory’ be longer or shorter? Why? What do you mean by stability?

b. Pick a new value for ‘time period for reconciling inventory’ that you feel will increase the stability of the system, and run the model. Do the results match your expectations?

c. Explain your results. Should management be more or less aggressive in managing inventories?

Model 3



The firm wants to know whether they should press for the right to schedule overtime/undertime in upcoming contract negotiations with the union.

The normal workweek is 40 hours. Preliminary negotiations with the union reveal that they would be willing to allow a minimum of 35 hours per week and a maximum of 50 hours, at time-and-a-half, in return for increase in employee benefits and pension fund contributions. You agree to evaluate the benefits of overtime/undertime with the model so management knows what overtime is worth in improved stability.

Schedule Pressure is the ratio of desired to real production rate and measures the need for overtime. The 'Effect of Schedule Pressure on Production' relates the overtime/undertime actually scheduled to the pressure for overtime/undertime.

(01) average productivity=20
Units: units/person/Week

(02) average order rate=SMOOTH(customer orders, time period for averaging orders)
Units: units/Week

(03) customer orders=1000*(1+STEP(0.1,10))
Units: units/Week

(04) desired inventory=average order rate*time period for inventory holdings
Units: units

(05) desired production=average order rate+ inventory correction
Units: units/Week

(06) desired workforce=desired production/average productivity
Units: person

(07) effect of schedule pressure=table1(schedule pressure)
Units: Dmnl (dimensionless)
The Effect of Schedule Pressure on Production relates the over / undertime actually scheduled to the pressure for over / undertime.

(08) hire rate=quit rate+workforce correction
Units: person/Week

(09) Inventory= production-shipment
initial value: 4000
Units: units

(10) inventory correction= (desired inventory-Inventory)/time period for reconciling inventory
Units: units/Week

(11) production=production rate*effect of schedule pressure
Units: units/Week

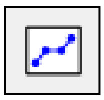
(12) production rate=Workforce*average productivity
Units: units/Week

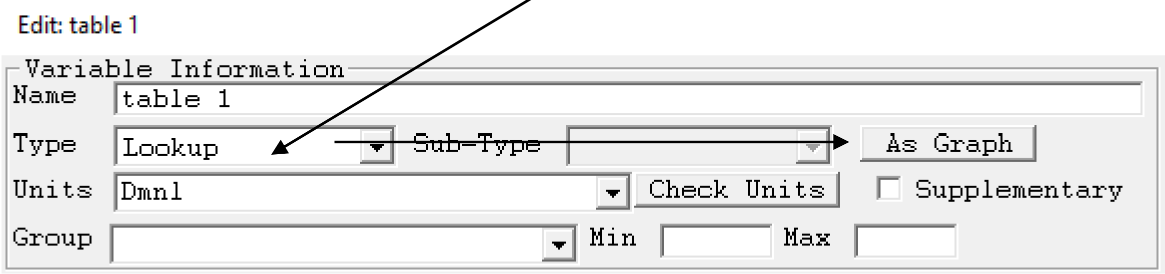
(13) quit rate=Workforce/time period of average employment
Units: person/Week

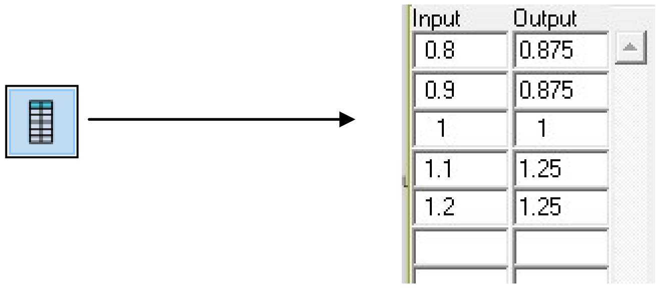
(14) schedule pressure=desired production/production rate
Units: Dmnl (dimensionless)
Schedule pressure is the ratio of desired to real production rate and measures the need for overtime.

(15) shipment=customer orders
Units: units/Week

(16) table1 = (0.8,0.875),(0.9,0.875),(1,1),(1.1,1.25),(1.2,1.25)
Units: Dmnl (dimensionless)







In a Lookup each row represents a point, this is a value for the "Output" variable that we know it takes for a value of the "Input" variable. For example if we know that when Price=3 the Sales are 120, we define the point (3,120)

(17) time period for averaging orders= 8
Units: Week

(18) time period for inventory holdings= 4
Units: Week

(19) time period for reconciling inventory= 8
Units: Week

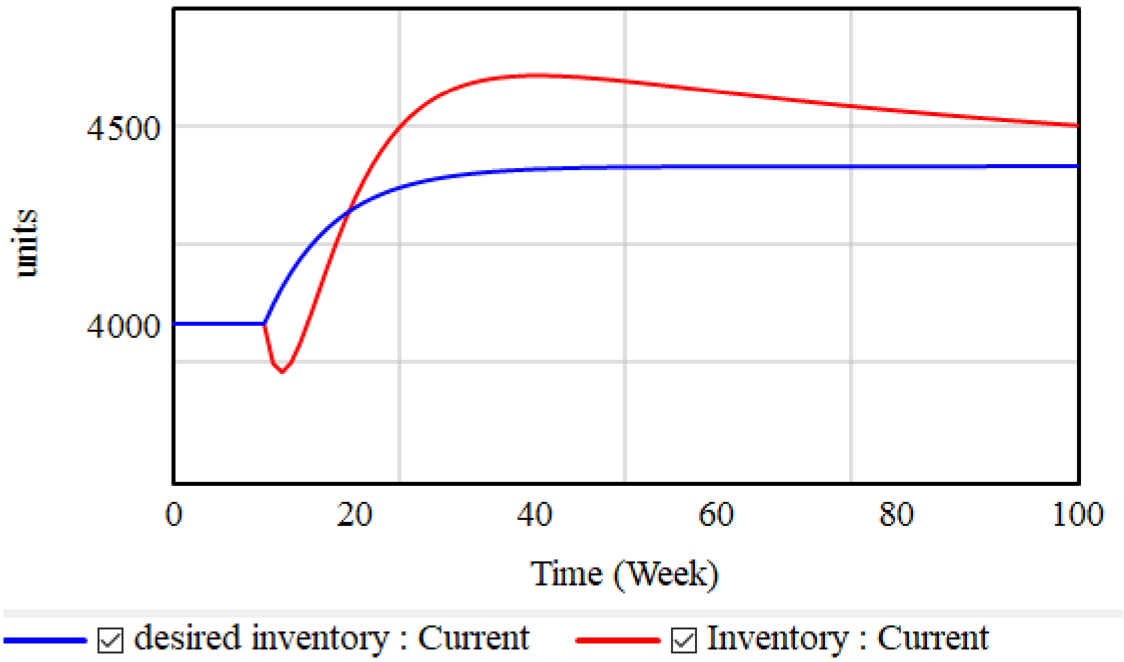
(20) time period of average employment= 50
Units: Week
Average duration of employment

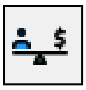
(21) time period to hire new workers=24
Units: Week
Time to correct labor

(22) Workforce= hire rate-quit rate
Initial value: 50
Units: person

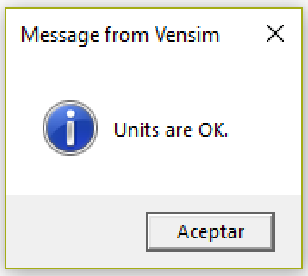
(23) workforce correction=(desired workforce-Workforce)/time period to hire new workers
Units: person/Week

Inherit scale	Variable	Dataset	Label	Line thickness	Pattern	Color	Units	Y-min	Y-max
	desired inventory			2	Solid line			3600	4800
☑	Inventory			2	Solid line				





Click the icon Units check, it must appear the message "Units are OK"



Questions

a. What does it mean when 'effect of schedule pressure’=1 ?

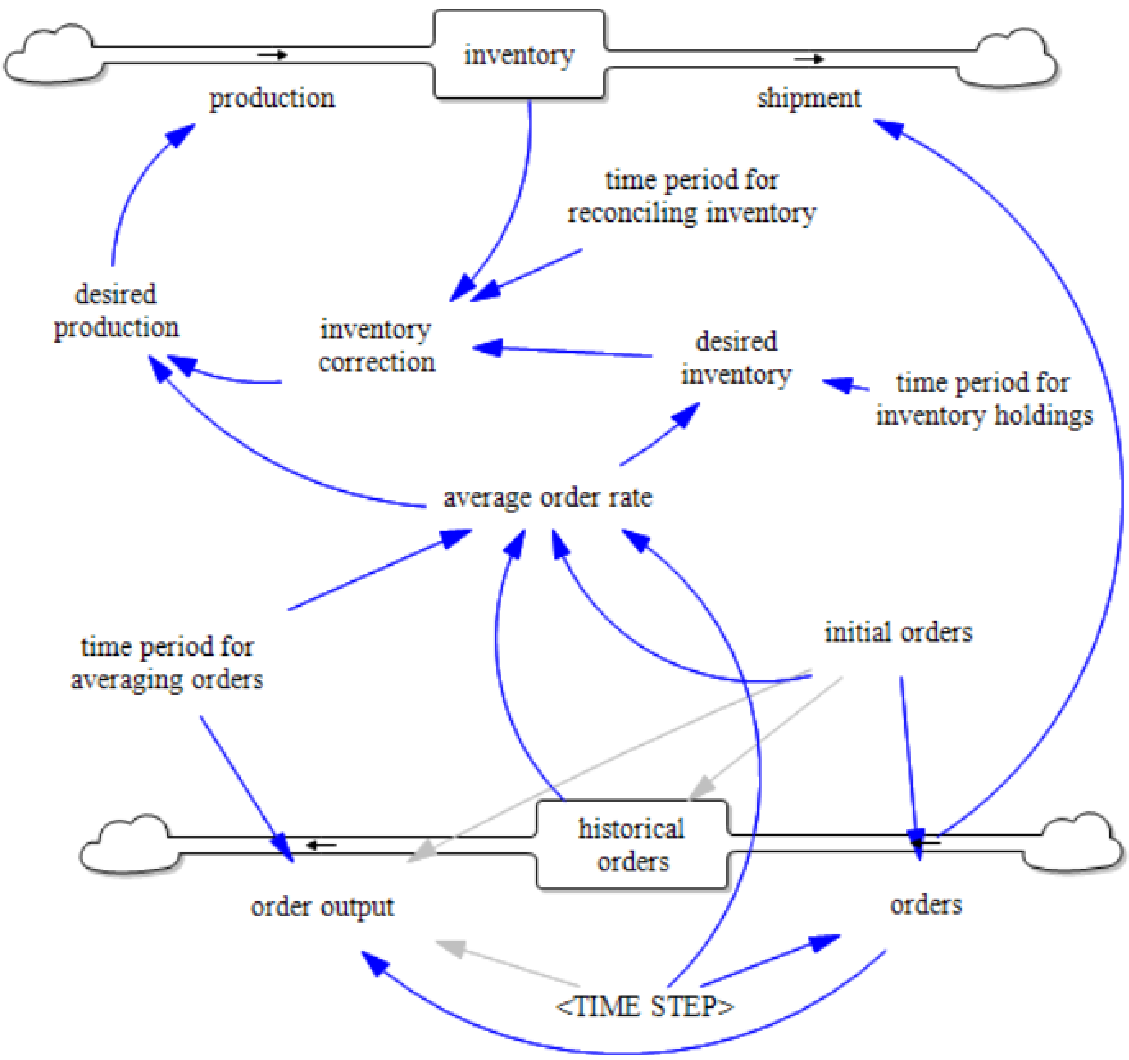
b. What is the effect of over / undertime on stability?

ANNEX

Average order rate

We can calculate the average order rate using a simple moving average (SMA), which is the simple average of the previous data points, instead of an exponential moving average (EMA), which is a first-order response that applies weight factors that decrease exponentially, as we did using the SMOOTH function.

To use an SMA the new model 1 introduces some changes (in black). Orders are stored for the order averaging time period and are used to calculate the average order rate. Variable order output uses the DELAY FIXED function that takes the order values after the time period to average the orders.



Equations

(01) average order rate=(initial orders/TIME STEP)+((historical orders-initial orders)/time period for averaging orders)
Units: units/Week

(02) desired inventory=average order rate*time period for inventory holdings
Units: units

(03) desired production=average order rate+inventory correction
Units: units/Week

(04) historical orders= INTEG (orders-order output
Initial value: initial orders
Units: units

(05) initial orders= 1000
Units: units

(06) inventory= INTEG (production-shipment
Initial value: 4000
Units: units

(07) inventory correction=(desired inventory-inventory)/time period for reconciling inventory
Units: units/Week

(08) order output= DELAY FIXED(orders, time period for averaging orders, initial orders/TIME STEP)
Units: units/Week

(09) orders=(initial orders+STEP(100,10))/TIME STEP
Units: units/Week

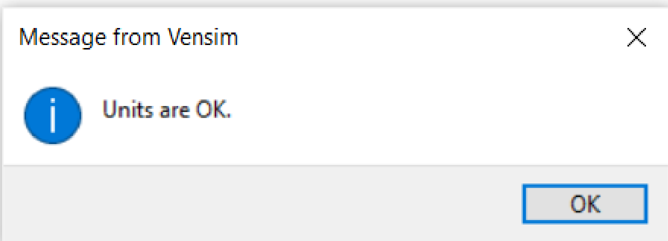
(10) production=desired production
Units: units/Week

(11) shipment= orders
Units: units/Week

(12) time period for averaging orders=4
Units: Week

(13) time period for inventory holdings=4
Units: Week

(14) time period for reconciling inventory=8
Units: Week



Simulation

The graph shows (in blue) the simulation using the SMOOTH function, and the simulation (in red) using the DELAY FIXED function that allows to calculate the simple moving average (SMA).